



Walchand Institute of Technology, Solapur
(An Autonomous Institute)
Affiliated to
Punyashlok Ahilyadevi Holkar Solapur University,
Solapur
Choice Based Credit System (CBCS)
Structure and Syllabus
for
Multidisciplinary Minor Program in
Civil Engineering
offered by
Department of Civil Engineering
for the Batch Admitted in
2025-2026

Multidisciplinary Minor Program in Civil Engineering

Legends Used

L	Lecture Hours / week
T	Tutorial Hours / week
P	Practical Hours / week
FA	Formative Assessment
SA	Summative Assessment
ESE	End Semester Examination
ISE	In Semester Evaluation
ICA	Internal Continuous Assessment
POE	Practical and Oral Exam
OE	Oral Exam
MOOC	Massive Open Online Course
HSS	Humanity and Social Science
NPTEL	National Programme on Technology Enhanced Learning
F.Y.	First Year
S.Y.	Second Year
T.Y.	Third Year
B. Tech.	Bachelor of Technology



Multidisciplinary Minor Program in Civil Engineering

Course Code Format									
2	5	C	E	U/P	2	C	C	1	T/L
Year of Syllabus revision	Program Code			U-Under Graduate P-Post Graduate	Semester No./ Year1/2/3/...8	CourseType		Course Serial No1-9	T-Theory, L-Lab session A-Tutorial P- Programming / Design

Program Code	
CE	Civil Engineering
Course Type	
BS	Basic Science
ES	Engineering Science
HU	Humanities & Social Science
MC	Mandatory Course
CC	Programme Core Compulsory Course
SN*	Self-Learning (N* indicates the serial number of electives offered in the respective category)
EN*	Programme Core Elective Course (N* indicates the serial number of electives offered in the respective category)
SK	Skill-Based Course
SM	Seminar
MP	Mini project
PR	Project
IN	Internship
ON*	Open Elective (N* indicates the serial number of electives offered in the respective category)
MD	Multidisciplinary Minor
EM	Entrepreneurship/Economics/Management Courses
FP	Community Engagement Project / Field Project



AE	Ability Enhancement Course
VE	Value Education Course
IK	Indian Knowledge System
VS	Vocational & Skill Enhancement Course
RM	Research Methodology
HN	Honors' Degree Course
HR	Honors Research

Sample Course Code	
25CEU3MD6T	Multidisciplinary Minor I



MDM Offered by Department of Civil Engineering

Semester	Course Code	Name of Course	Engagement Hours			Credits	SA		FA		Total
			L	T	P		Theory	OE/POE	ISE	ICA	
III	25CEU3MD6T	Multidisciplinary Minor- I Smart Buildings	2	-	-	2	60	-	40	-	100
IV	25CEU4MD6T	Multidisciplinary Minor- II Geoinformatics	1	-	-	1	-	-	50	-	50
V	24CEU5MD6T	Multidisciplinary Minor-III Environmental Impact Assessment	3	-	-	3	60	-	40	-	100
VI	24CEU6MD6T	Multidisciplinary Minor -IV Infrastructural Systems	2	-	-	2	60	-	40	-	100
VII	23CEU7MD6T	Multidisciplinary Minor -V Disaster preparedness and planning	3	-	-	3	60	-	40	-	100
VII	23CEU7MD6A	Multidisciplinary Minor-V Disaster preparedness and planning	-	1	-	1	-	---		25	25
Subtotal			11	1		12	240	-	210	25	475
Laboratory Courses											
IV	25CUE4MD6L	Multidisciplinary Minor- II Geoinformatics Lab	-	-	2	1	-	-		25	25
VI	24CEU6MD6L	Multidisciplinary Minor -IV Infrastructural Systems Lab	-	-	2	1	-	-		25	25
Subtotal			11	-	4	2	-	-		50	50
Grand Total			11	1	4	14	240	-	210	75	525



WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR

(An Autonomous Institute)

Second Year B.Tech. (Multidisciplinary Minor in Civil Engineering), Semester-III

25CEU3MD6T: Multidisciplinary Minor I-Smart Buildings

Teaching Scheme		Examination Scheme	
Lectures	2 Hours/week	ESE	60 Marks
Credits	2	ISE	40 Marks
Introduction:			
This course provides a thorough introduction to Smart Building Technology and its concepts related to building Management Systems for the development of intelligent and smart buildings.			
Course Prerequisite:			
Basic understanding of various building units and their functions			
Course Objectives:			
1. To enhance the knowledge of Smart Building Technologies 2. To know about the concept of Building Automation 3. To evaluate the functioning of Smart Buildings. 4. To analyze the integration of Integrated Building Management Systems (IBMS) and IoT in smart buildings. 5. To describe the application of smart technologies in HVAC, lighting, fire safety, and security systems.			
Course Outcomes:			
After completing the course, students will be able to, 1. Recognize the system of Smart Technology 2. Identify the technology involved in the Integrated Building Management System (IBMS) in Smart buildings 3. Illustrate the philosophy of building Management systems and IoT. 4. Comprehend the technology available in Heating, Ventilation, and Air Conditioning (HVAC) and Lighting control systems. 5. Describe technology available in fire safety and security systems for intelligent buildings.			
Unit – I	Introduction to Smart Buildings		2 Hours
Definition and Overview, History and evolution, Benefits and challenges			



Unit – II	Integrated Building Management System (IBMS) in Smart Buildings	5 Hours
Introduction to Integrated Building Management System (IBMS) Smart buildings. Components of Smart Building IBMS.		
Unit – III	Building Management Systems (BMS) and IoT	5 Hours
Building Management Systems (BMS): Components of BMS, Functions and Architecture. Internet of Things (IoT) in Smart Buildings: IoT fundamentals: IoT devices and sensors, Communication protocols.		
Unit – IV	Heating, Ventilation, and Air Conditioning (HVAC) and Lighting Controls Systems:	6 Hours
Human Comfort: Air Quality-Thermal comfort in buildings-Ventilation-Classification of HVAC - DDC applications - Control Panel: HVAC Control Panel, Networks, BAC Net, Modbus, LON-Electrical Installation Power Transmission -Smart Lighting Systems		
Unit – V	Fire Protection Systems:	6 Hours
Introduction: Causes and Stages of Fire, Fire Service Installation, Fire Alarm. Principles of Operation, Fire Sensors: Smoke detectors and their types, Fire control panels, Fire and Life safety, and Fire Management in buildings.		
Unit – VI	Security Systems	6 Hours
Introduction to Security Systems: Concepts, Perimeter Intrusion, Security Design, Access control system, RFID-enabled access control, Computer system access control: CCTV, Components of CCTV system, Central alarm systems, and Structural health monitoring systems.		
INTERNAL CONTINUOUS ASSESSMENT (ICA)		
Internal Continuous Assessment (ICA) shall consist of a minimum of five assignments based on the entire curriculum, which will be assessed by the concerned faculty.		
Text Books		
1. Shengwei Wang, “Intelligent Buildings and Building Automation”, Spon Press, London, 2009. 2. Derek Clements and Croome, “Intelligent Buildings: An Introduction”, Routledge, 2013, Civil Engineering.		
Reference Books		
1. Derek Clements and Croome, “Intelligent Buildings: An Introduction”, Routledge, 2013 2. James Sinopoli “Advanced Technology for Smart Buildings”, Artech House, 2016 3. James.M.Sinopoli, Smart Buildings Systems for Architects, Owners and Builders, Publishers:” Butter worth Heinemann, 2009. 4. James Kachadorian “Passive Solar House: the complete Guide to Heating and Cooling Your Home “ Chelsa Green Publishing: Revised and expanded Second edition, 2006 Ron Bakker, “Smart Buildings: Technology and the Design of the Built Environment”, RIBA Publishing, 2020.		
e-Resources		
1. NPTEL (India): Introduction to Smart Grid-https://nptel.ac.in 2. edX: Smart Buildings and the Internet of Things (offered by TU Delft)-https://www.edx.org		





WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR

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Second Year B.Tech. (Multidisciplinary Minor in Civil Engineering), Semester-IV

25CEU4MD6T: Multidisciplinary Minor II-Geoinformatics

Teaching Scheme		Examination Scheme	
Lectures	1 Hour/week	ISE	50 Marks
Practicals	2 Hours/week	ICA	25 Marks
Credits	2		

Introduction:

This course covers the fundamentals of remote sensing, GIS, GPS, and geoinformatics applications essential for spatial data analysis. Topics include remote sensing principles, GIS components and data structures, GPS technology and geo-positioning techniques, and geoinformatics applications in natural resource management, environmental studies, and urban planning, emphasizing practical applications in engineering and environmental practice.

Course Prerequisite:

Basic knowledge of Physics, Geography, and Earth Sciences

Course Objectives:

1. Describe remote sensing history, EMR interaction, platforms, sensors, and resolution types for analysing satellite imagery data.
2. Examine GIS components, spatial and attribute data characteristics, raster vs. vector structures, and analyze satellite data.
3. Analyze GPS history, segments, geo-positioning techniques, advantages, limitations, and practical applications in various fields.
4. Utilize geoinformatics for managing natural resources, environment, disasters, utilities, urban studies, military, navigation, and agriculture problems.

Course Outcomes:

After completing the course, students will be able to

1. Describe remote sensing history, concepts, EMR interaction, platforms, sensors, and types of resolution in satellite imagery analysis.
2. Analyze and apply knowledge of GIS components, spatial and attribute data, raster vs. vector structures, and process satellite data in GIS.
3. Analyze GPS history, advantages, limitations, segments, geo-positioning techniques, and practical applications across various fields.
4. Proficient to apply geoinformatics to manage natural resources, environment, disasters,



utilities, urban areas, military, navigation, and agriculture, solving related problems.		
Unit – I	Introduction to Remote Sensing	4 Hours
Remote sensing history, concept and principles, Interaction of EMR with atmosphere and Earth's surface, Remote sensing platforms and sensors, Concept of Resolution-spatial, spectral, temporal and radiometric, Satellite and their characteristics- geostationary and sun-synchronous		
Unit – II	Fundamentals of Geographic Information System:	4 Hours
Components of GIS, Data used in GIS, characteristics of Spatial Data – sources of spatial and attribute data, data structure - raster and vector, Processing of various satellite data.		
Unit – III	Global Positioning System and its Applications	4 Hours
History of GPS, Advantages and limitations of GPS, Segments of GPS - Control segment, Space segment, User segment, Geo positioning - Point positioning, Relative Positioning, Static Positioning, Kinematics Positioning, Uses of GPS.		
Unit – IV	Application of Geoinformatics:	3 Hours
Application of Geoinformatics in Natural Resources Management, Environmental Studies, Disaster Management, Utilities Management, Urban Studies, Military Applications – Navigation - Location Based Services and Agriculture.		
INTERNAL CONTINUOUS ASSESSMENT (ICA)		
Internal Continuous Assessment (ICA) shall consist of a minimum of five assignments based on the entire curriculum and which will be assessed by the concerned faculty.		
Text Books		
1. Shengwei Wang, "Intelligent Buildings and Building Automation", Spon Press, London, 2009. 2. Derek Clements and Croome, "Intelligent Buildings: An Introduction", Routledge, 2013, Civil Engineering.		
Reference Books		
1. Derek Clements and Croome, "Intelligent Buildings: An Introduction", Routledge, 2013 2. James Sinopoli, "Advanced Technology for Smart Buildings", Artech House, 2016 3. James.M.Sinopoli, Smart Buildings Systems for Architects, Owners and Builders, Publishers:" Butterworth Heinemann, 2009. 4. James Kachadorian, "Passive Solar House: The Complete Guide to Heating and Cooling Your Home, "Chelsea Green Publishing: Revised and expanded Second edition, 2006. Ron Bakker, "Smart Buildings: Technology and the Design of the Built Environment", RIBA Publishing, 2020.		



5. A Course in Civil Engineering Drawing, Sikka, S.K., Kataria & Sons
6. Engineering Drawing, Dhanarajay A. Jolhe, Tata McGraw-Hill

e-Resources

1. NPTEL-Geoinformatics for Natural Resource Management by IIT Roorkee-nptel.ac.in
2. Coursera-GIS, Mapping, and Spatial Analysis Specialization (UC Davis)-coursera.org





WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR

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Third Year B.Tech. (Multidisciplinary Minor in Civil Engineering), Semester-V

24CEU5MD6T: Multidisciplinary Minor III- Environmental Impact Assessment

Teaching Scheme		Examination Scheme	
Lectures	3 Hours/week	ESE	60 Marks
Practical	-	ISE	40 Marks
Credits	3	ICA	-

Introduction:

This course covers the principles, methodologies, and processes of Environmental Impact Assessment (EIA), focusing on evaluating environmental impacts, mitigation strategies, and regulatory compliance in real-world development projects.

Course Prerequisite:

Basics of Environmental Engineering

Course Objectives:

1. To introduce students to the concept and importance of Environmental Impact Assessment in sustainable development.
2. To provide knowledge on the methodologies used for EIA and the preparation of Environmental Impact Statements (EIS).
3. To familiarize students with the legal and regulatory framework for EIA in various countries.
4. To enable students to analyze the environmental impacts of different industrial, infrastructural, and developmental projects.

Course Outcomes:

After completing the course, students will be able to

1. Describe the principles, objectives, and importance of EIA in environmental management.
2. Identify key components of an Environmental Impact Assessment and the methodology used to conduct assessments.
3. Evaluate the environmental impacts of various development projects.
4. Prepare Environmental Impact Statements (EIS) and suggest mitigation measures for identified impacts.



Unit – I	Introduction to EIA	6 Hours
Definition and Scope of EIA, Importance of EIA in sustainable development, Evolution and historical perspective of EIA, Types of EIA: Project-specific and Strategic Environmental Assessment (SEA), EIA Process and Methodology		
Unit – II	EIA Methodologies	6 Hours
Screening, Scoping, and Public Participation, Baseline Data Collection and Impact Prediction, Tools and Techniques for Impact Assessment: Environmental Indicators, Mathematical Modelling for Impact Prediction, GIS and Remote Sensing in EIA, Environmental Impact Statement (EIS) preparation		
Unit – III	Environmental Impact Evaluation	8 Hours
Identifying significant impacts, Impact classification: Positive vs. Negative, Direct vs. Indirect, Social, Economic, and Cultural Impact Assessment, Quantitative vs. Qualitative Assessment, Impact Significance Criteria, Risk Assessment in EIA		
Unit – IV	EIA for Different Sectors	8 Hours
Industrial Projects: Manufacturing, Mining, etc., Infrastructure Projects: Highways, Dams, Airports, etc., Urban Development and Land Use Change, Environmental Impacts of Agriculture, Forestry, and Fisheries, Case Studies of sector-specific EIA		
Unit – V	Legal and Regulatory Framework	6 Hours
National and International EIA Regulations, Role of Government Agencies and NGOs in EIA, Environmental Laws and Policies, Environmental Clearance Process, EIA in Developing Countries and Challenges		
Unit – VI	Mitigation and Monitoring	6 Hours
Mitigation Strategies for Environmental Impacts, Environmental Management Plan (EMP), Post-EIA Monitoring and Evaluation, Case Study of EIA in practice: Successful EIA projects and failures		
The ICA includes at least 6 assignments, where students are required to develop Environmental Impact Assessment (EIA) reports and related applications for real-world development projects		
Text Books		
1. Morrison-Saunders, A., Advanced Introduction to Environmental Impact Assessment, Edward Elgar Publishing, 2 nd Edition, 2023.		



- Islam, K. M. B., & Nomani, Z. M. (Eds.), Environment Impact Assessment: Precept & Practice, CRC Press, 1st Edition, 2022. - Bhateria, R., Sharma, M., Singh, R., & Kumar, S., Environmental Impact Assessment: A Journey to Sustainable Development, Springer, 1st Edition, 2024. - Rosales, J., Environmental Impact Assessment, Delve Publishing, 1st Edition, 2021.
Reference Books
- Fonseca, A. (Ed.), Handbook of Environmental Impact Assessment, Edward Elgar Publishing, 1st Edition, 2022. - Hanna, K. (Ed.), Routledge Handbook of Environmental Impact Assessment, Routledge, 1st Edition, 2022. - Hosetti, B. B., & Kumar, A., Environmental Impact Assessment and Management, Daya Publishing House, 1st Edition, 2024. - Carroll, B., Environmental Impact Assessment Handbook: A Practical Guide for Planners, Developers and Communities, Emerald Publishing Limited, 3rd Edition, 2019.
e-Resources
- NPTEL Course: Environmental Impact Assessment - Coursera Course: Introduction to Environmental Impact Assessment





WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR

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Third Year B.Tech. (Multidisciplinary Minor in Civil Engineering), Semester-VI

24CEU6MD6T: Multidisciplinary Minor IV-Infrastructural Systems

Teaching Scheme		Examination Scheme	
Lectures	2 Hours/week	ESE	60 Marks
Practical	2 Hours/week	ISE	40 Marks
Credits	3	ICA	25 Marks

Introduction:

This course provides a comprehensive understanding of the planning, design, construction, and management of critical infrastructure systems, including transportation, water supply, energy, telecommunications, and waste management.

Course Prerequisite:

Basic understanding of civil engineering principles, Environmental science, and Surveying and Geomatics

Course Objectives:

1. Understand the key components and interdependencies of infrastructural systems.
2. To examine the components and technologies involved in transportation, water, and wastewater infrastructure.
3. To explore the concept, objectives, and development frameworks of smart cities.
4. To analyse the planning, integration, and sustainability of smart city infrastructure, including energy, housing, and safety systems.
5. To evaluate and apply digital and intelligent tools such as GIS, GPS, and ITS for managing urban mobility and infrastructure.

Course Outcomes:

After completing the course, students will be able to

1. Describe various types of infrastructure systems and their relevance to socio-economic development.
2. Differentiate and illustrate the structural and functional aspects of transportation and water/wastewater systems.
3. Describe the concept and evolution of smart cities, along with the associated challenges and goals.
4. Analyse the integration of sustainable practices into smart city infrastructure, such as energy systems, green buildings, and disaster management.
5. Apply smart infrastructure tools (GIS, ITS, GPS) to develop and manage transportation and mobility services.



Unit – I	Introduction to Infrastructure Systems	4 Hours
Definition and importance of infrastructure, Classification of infrastructure (transportation, water, energy, etc.), Overview of urban and rural infrastructure, Stakeholders in infrastructure development, and Infrastructure planning process and regulatory framework		
Unit – II	Transportation Infrastructure	4 Hours
Roadways, railways, airports, and ports, Urban transportation systems (metro, BRT, etc.), Intelligent Transportation Systems (ITS), and Case studies of transportation projects		
Unit – III	Water and Wastewater Infrastructure	4 Hours
Water supply systems (sources, treatment, distribution), Wastewater management (collection, treatment, disposal), Stormwater drainage and flood control, and Integration with urban planning and sustainability aspects		
Unit – IV	Fundamentals of Smart City and Infrastructure	6 Hours
Introduction of Smart City, Concept of smart city, Objective for smart cities, History of Smart City World and India. Need to develop a smart city, Challenges of managing infrastructure in India and the world, various types of infrastructure systems, Infrastructure needs assessment		
Unit – V	Planning and development of Smart city Infrastructure	6 Hours
Energy and ecology, solar energy for smart city, Housing, sustainable green building, safety, security, disaster management, economy, cyber security, Project management		
Unit – VI	Tools for Smart City Infrastructure	6 Hour
Intelligent Transport Systems, Smart Vehicles and Fuels, GIS, GPS, Navigation Systems, Traffic Safety Management, Mobility Services, E-ticketing		
ICA consists of a minimum of 06 assignments based on the above syllabus		
Text Books		
1. Grigg, N. S. Infrastructure Engineering and Management, McGraw Hill Education, Latest Edition. (Comprehensive coverage of infrastructure systems with planning and management aspects) 2. Thagesen, B. Highway and Traffic Engineering, Thomas Telford Publishing. (For transportation infrastructure details) 3. Moser, D. A., & Markus, M. D. Engineering Economic Analysis of Water Supply Systems, American Water Works Association. (Water infrastructure economics and planning). 4. Infrastructure: A Guide to the Industrial Landscape – Brian Hayes 5. Civil Infrastructure Systems: Intelligent Renewal – Hojjat Adeli 6. Sustainable Infrastructure: Principles into Practice – Charles A. Reed & Brian Robert		
Reference Books		



1. Sullivan, A. M., & Sheffrin, S. M. <i>Economics: Principles in Action</i>, Pearson Education. 2. Hansen, W. G. <i>Urban Planning and Infrastructure</i>, MIT Press. 3. Patra, S. C. <i>Municipal Engineering: Principles and Practice</i>, Narosa Publishing.
e-Resources
1. www.worldbank.org 2. www.asce.org 3. https://elibrary.worldbank.org 4. https://smartcities.gov.in





WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
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Third Year B.Tech. (Multidisciplinary Minor in Civil Engineering), Semester-VI

23CEU7MD6T: Multidisciplinary Minor V-Disaster Preparedness and Planning

Teaching Scheme		Examination Scheme	
Lectures	3 Hours/week	ESE	60 Marks
Tutorial	1 Hours/week	ISE	40 Marks
Credits	3	ICA	25 Marks
Introduction:			
This course provides a comprehensive understanding of disaster preparedness, planning, risk reduction strategies, and management practices to minimize the impacts of natural and manmade disasters.			
Course Prerequisite:			
Basic knowledge of civil engineering, environmental science, and natural hazard processes.			
Course Objectives:			
1. To understand fundamental concepts and terminologies of disaster management, including hazard, vulnerability, risk, resilience, and mitigation. 2. To classify and describe various types of disasters, including natural and man-made disasters, along with their causes and characteristics. 3. To analyze the multidimensional impacts of disasters on environmental, social, economic, and human systems. 4. To impart knowledge of disaster risk reduction (DRR) principles and management practices, including disaster management cycle, early warning systems, and stakeholder roles. 5. To understand the interrelationship between disasters and development, and apply sustainable approaches for disaster-resilient planning and reconstruction			
Course Outcomes:			



After completing the course, students will be able to 1. Interpret key concepts, terminologies, and frameworks of disaster management. 2. Classify and evaluate different types of natural and man-made disasters based on their characteristics and impacts. 3. Analyze disaster impacts on various sectors, including environmental, social, economic, and health dimensions. 4. Apply disaster risk reduction strategies, including mitigation, preparedness, response, and recovery measures in real-world scenarios. 5. Develop disaster-resilient planning and sustainable development strategies, considering environmental and socio-economic factors.		
Unit – I	Introduction to Various Concepts	6 Hours
• Concepts and definitions: disaster, hazard, vulnerability, resilience, risks severity, frequency and details, capacity, impact, prevention, mitigation.		
Unit – II	Disasters and its types	9 Hours
• Natural disasters (floods, drought, cyclones, volcanoes, earthquakes, tsunami, landslides, coastal erosion, soil erosion, forest fires etc.); • Manmade disasters (industrial pollution, artificial flooding in urban areas, nuclear radiation, chemical spills, transportation accidents, terrorist strikes, etc.).		
Unit – III	Disaster Impacts	8 Hours
• Disaster impacts environmental, physical, social, ecological, economic, political, etc.); health, psycho-social issues; demographic aspects (gender, age, special needs); hazard locations; global and national disaster trends; climate change and urban disasters		
Unit – IV	Disaster Risk Reduction (DRR)	12 Hours
• Disaster Management Cycle – its phases: prevention, mitigation, preparedness, relief and recovery; structural and non-structural measures; risk analysis, vulnerability and capacity assessment; early warning systems, • Post-disaster environmental response (water, sanitation, food safety, waste management, disease control, security, communications); • Roles and responsibilities of government, community, local institutions, NGOs and other stakeholders; Policies and legislation for disaster risk reduction, DRR programmes in India • National Disaster Management Authority.		
Unit – V	Inter-Relationship Between Disasters and Development	10 Hours



Factors affecting vulnerability, such as impact of developmental projects and environmental modifications (including dams, land use changes, urbanization etc.), sustainable and environmentally friendly recovery, reconstruction and development methods.
ICA consists of a minimum of 05 assignments based on the above syllabus
Text Books
- Pradeep Sahni, Disaster Risk Reduction in South Asia, Prentice Hall of India, New Delhi, 2004. - B.K. Singh, Handbook of Disaster Management: Techniques and Guidelines, Rajat Publications, New Delhi, 2008. - G.K. Ghosh, Disaster Management, APH Publishing Corporation, New Delhi, 2006. K. K. Shrivastava, Disaster Management: Challenges and Solutions, CBS Publishers & Distributors, New Delhi.
Reference Books
- Coppola, D. P., Introduction to International Disaster Management, Butterworth-Heinemann, Elsevier. - Alexander, D., Principles of Emergency Planning and Management, Oxford University Press. - UNISDR, Global Assessment Report on Disaster Risk Reduction, United Nations Office for Disaster Risk Reduction. - National Disaster Management Authority (NDMA), Government of India – Guidelines and Publications. - Inter-Agency Standing Committee (IASC), Guidelines on Mental Health and Psychosocial Support in Emergency Settings, Geneva. - Emergency Medical Services Authority (EMSA), California, Disaster Medical Systems Guidelines, 2003. - Pradeep Sahni & Alka Dhameja, Disaster Mitigation: Experiences and Reflections, Prentice Hall of India.
e-Resources
- National Disaster Management Authority (NDMA): https://ndma.gov.in - Ministry of Home Affairs – Disaster Management Division: https://ndmindia.nic.in - United Nations Office for Disaster Risk Reduction (UNDRR): https://www.undrr.org - National Institute of Disaster Management (NIDM): https://nidm.gov.in

